ActInf Livestream #039.2 ~ "Morphogenesis as Bayesian inference...."

Livestream | 1:38:01

All right.

Hello, everyone.

Welcome to ActinFLAB livestream number 39.2.

It's March 9th, 2022.

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We're a participatory online lab that is communicating, learning, and practicing applied active inference.

You can find us at the links here on the slide.

This is a recorded and archived livestream, so please provide us with feedback so we can improve on our work.

All backgrounds and perspectives are welcome, and we'll be following video etiquette for livestreams.

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Today in stream number 39.2, we're in our third discussion around this paper,

Morphogenesis as Bayesian Inference, a Variational Approach to Pattern Formation and Control in

Complex Biological Systems, by Franz Kuchling, who's here with us, as well as Frustin, Georgiev, and Levin.

So we're going to have a fun discussion in the .2.

The .0 was some background and context, hopefully overviewing the paper.

The .1, we had a discussion that opened up several threads.

And today we will, in the .2, say hello on the introductions and then just jump in to a blank slide or to a not blank slide.

So, I'm Daniel.

I'm a researcher in California, and I'll pass it to Dean.

Morning, I'm Dean.

I'm in Calgary.

And I am going to say that I have one thing I want to look at down the road, which is how long is a person a stem cell?

Which I think this paper might help us parse a little bit, and I'll pass it down to Blue.

Hi, I'm Blue.

I'm a researcher in New Mexico.

And I also, similar to Dean, have questions about when a cell becomes a cancer cell and like, how does it develop its own generative model?

And just in general, like when you're looking at complex systems, how these small perturbations can lead to like a totally different effect downstream, which I think is kind of related to the stem cell question that Dean had.

And I'll pass it over to the first author, Franz.
Hi, I'm Franz.
I am a PhD student in the Levin lab in Tufts, Medfield, near Boston.
And yeah, I study part of active inference in my theoretical work, and I do a lot of experimental
work as well, looking at information processing in organisms and cells.
Cool.
So where does stemness, or how do stem cells play into this paper and this line of research, Franz?
Definitely something we've touched on, but just to kind of start generally,
and then we'll zoom in somewhere. So take it away.
Sure. The idea is that the cells initially, when they're in the simulation, when they're
being initialized, they have the random initialized and they have very low beliefs about what cell
type they are. They can be one of four different cell types in this particular simulation. And they all
share a model, a tallyt morphology they could achieve that they essentially are programmed to achieve at some point.
But none of the cells are any of the given cell types initially, at least not with any
significant precision. And as they move along, signaling to each other and sensing each other and secreting
signal types in response to each other, they then slowly by slowly infer a cell type, which essentially
is differentiation.
So cells starting with low precision about what kind of cell they believe themselves to be. And then
there's a process of differentiation. So Diener Blue, any thoughts on that sort of stem to differentiated
continuum? And I'll pull up an image from a non-active perspective that also shows that kind of a
continuum.
So I have just something to add that ties into the paper where there's aberrant signaling. So
a stem cell differentiates, but then during the process of carcinogenesis, like retains some kind
of, or D differentiates into something that's more similar to a stem cell than it is to a different
kind of cell. And so this plasticity is important to retain, like, so that we can repair ourselves,
but also it can lead to kind of an aberrant signaling pattern.
Yeah.
Yeah.
Like this notion of having a terminal cell
revert to a state and regain the ability of having stemness is one of the characteristics of cancer,
because it can then sometimes differentiate into other cell types that are like normal and downstream
of a precursor or abnormal in some other way.
How is this...
And it's also something...
Yeah, go ahead, friends.
I was going to say, it's also something that happens in, especially in higher organisms,
right? Not every organism keeps absolute stem cells in all kinds of populations. One example that I used

to work on is in zebrafish, where you have an always developing eye and there's certain...
There are parts of the retin, essentially, that can de-differentiate some extents. They're not entirely stem... They have a stem cell niche as well, but there's also, especially in higher organisms, while they keep those populations of either stem cells or of already partially differentiated cells that can de-differentiate. So it's an important part in regeneration. And ironically, this is something that's not been studied for a while, but there's older research that was done where highly regenerative organisms, when they undergo... They did it as in salamanders, and they injected them with carcinogens, and they basically had it... You know, saw a tumor progressing, and they cut up a limb, and these are highly regenerating. We actually re-normalize the tumors, essentially.
So initiation of regeneration in those organisms seems to actually counteract cancer progression, and even kind of remove the tumors, which is very interesting. And it kind of... This feeds a little bit into the idea in this paper as well that regeneration, right? What happens is you certainly now have a much higher flow of information, things need to rearrange to re-differentiate, and that is kind of like what I was thinking as well with the simulation here of the cancer rescue phenomena, where you rescue this formation of aberrant cells and aberrant signaling by having a stronger contact between cells as well. I'm not saying I'm modeling generation here, although you can... I think in the original paper, Carl has done some way to cut, I think, part off and let it grow again. I think that has been done in the original paper in the annoying one's place. But that's something to kind of keep in mind. This is something that is experimentally highly studied to an extent.
So one general note, and then a question. This idea that the organisms that have the highly regenerative capacity, and there's a continuum from regrowing just the tail to regrowing a whole limb to kind of an extreme or a highly regenerative case like the planaria that was studied in this paper. So it's all kind of on a continuum from total ability to regenerate from any adult somatic cell can fully recapitulate the full population all the way to a total opposite of that. And then you pointed out that it might be related to cancer progressions and phenotype. And that speaks to what Blue brought up about the like system level effects. There's the individual cell and then there's the system level, because clearly there's something about the system level in the regenerative situation that would lead to this being different. Or maybe it could be explained by just something internal to the model. Anyways, you mentioned that higher information was associated with rearrangement. Could you explain a little bit about that? Or how does it play out in this paper or in general?
Yeah, it's basically the idea that it's not so much the total information content, but the flow of it.
And I'm talking a little bit about this last time, right? It's hard to really kind of quantify this in a more meaningful biological sense. But the idea is that it's information flow. Like I think Dean asked this question, like is more information always good? And my kind of answer to that was, it's within context of what the participating organisms can receive and can process. So in the sense that if you have cells that are processing certain information with a certain kind of rates, time constants, then kind of getting information in within with that timeframe in mind is what makes it more likely for them to kind of understand better their environment and react to it, as opposed to a cell,

you can think of it also cancer cells that, you know, have the whole sensory part completely perturbed,
that do not react to the environment anymore, then everything else becomes noise and it gets ignored.
That's kind of the idea. So there is, there is a lot of different differences in how the cancer cells,
because they have changed ion profile, ion channel profiles, they have changed
machinery of what they are actually transcribing. So all of that leads to the difference in how signals
are being processed. And understanding that and then reacting to that is, I think, is a key to
to understanding and treating this disease.

Dr. David Pérez

Awesome. Thanks, Dean.

I don't know if I have the answer to my question, which is how long is a person,
a, in finger, in scare quotes, a stem cell. But one of the things you mentioned,
Franz, in the point one was, I think Stephen asked the question about how much volume in the flow. And what
I've heard you say back was, it's always a state space dependency, which, which I, when I'm reading
the paper, it was, I think you made a really good sound case for that. But then one of the things that
when you had to, when you had to sign off, one of the questions that I brought up, hopefully that maybe
we can discuss a little bit today is, they can figure four and five of the paper, all that black area
around when the sort of cells move into place. That's a crude way of describing what's going on
there. But all that black area around that, am I, is it fair to say that that's kind of the,
the state space dependency? Or is that just assumed? And how would we, because Daniel did a good job of
sort of saying, there's some invisibility aspects to this when we do multiple layering. And so,
how, how do you give people a sense of what's going on in that, in that black, that blacked out area?
Right? I think, I think black is a good, a good color representation, because it's not really
a color, I don't think. But it, but it kind of speaks to this idea that we've got these low beliefs,
but how long, or how long as a person, or as cells, as a cluster of cells, is that beneficial?
Right? When we're talking, we're talking of course here on under the big umbrella of morphology. So
that was kind of where I was wondering today, maybe you could, you could explain it, because you, the
focus, the eye tends to focus on the object within the frame, as opposed to sort of the area around the,
what we would consider to be the object. So maybe you can kind of help me understand this state space
dependency piece. Yeah, I mean, so if you have the look at figure two, so, and what I didn't plot in
the figure four and five, but in two I did, is the back of concentrations, right? So there are, of course,
some, some levels of constant, of signaling molecules in the background as well, which will also,
of course, be the case in the, in the actual simulations, down the line of four and five.
The, the blackness, so the one interesting part about the simulation, which was done mainly for
simplicity and lack of necessity, is to not actually have an external environment per se,
like there's nothing, right? There's no external force, like, outside of base, everything in the
simulations is done by the cells, where the cells, they make the environment, right? So there is,
essentially there's an external environment by the signaling molecules that they put out,

but we don't have any external, limited, like, limits around it. We just basically, that's just basically plot, the signaling molecules don't go faster than that. And there isn't any external force, which we could do, right? We, we, the power of the simulations, where you do this asymmetry, we essentially get, we get sort of an, an implicit environment by changing the response to the signals around, which you can imagine that you put something in the, in the medium that, that flips how, how the signal is being processed, and then that's it. So back to your question. So this, about this blackness, and when is that beneficial, if I answered that question correctly, it's, there is, of course, like a hype, like a precision component to it and kind of more, less hand, less hand wavy. If, you know, if, if your environment is very volatile, and constantly changes, and you keep, right, you have a, not, not this, this is a closed system, right? If you had an open system, you constantly would get signals also from the environment, this will also become all a lot messier, and especially initially, right? In the, in the first frames of that figure, four and five, earlier the simulations, they all started together. I mentioned last time, there are certain smoothness requirements where like, if, if we had to put this dampening on the sensitivity a long time, so inverse dampening, it will get more sensitive with time, which is because initially they're very unstructured, they have, you know, very low power beliefs, and they will jump crazy if they have high sensitivity to each other, because they're in such close proximity. And
that is something that we, of course, do see in biology, right? We do be most, especially the higher organisms, right? They start off usually in close environments. Now, of course, you can, you can argue this, you don't need to go to any information stuff, you can just argue, well, it's just, you know, thermally insulated, they have their own food supplies, it makes, you know, just makes like a more safe zone, like just physically, energy speaking. But I do think there's a big component to it, it's like shutting this highly volatile, you know, itself inferring organism off from influence from the environment at a stage when all these cells are figuring out what to do each other.
So they need, they kind of need to have a kind of a barrier between signals from the environment, which normally, you know, to us adults, to a mature organism are not really dangerous, but to end developing organisms full of stem cells can be problematic and will basically would would perturb its natural kind of progression in this state space on this sulfate acquisition.
There is a cool paper I can reference to you from Chris Fields, who I am from disclosure, I'm working with and very much admire him, who wrote a paper about multicellularity as exactly the consequence of what I'm talking about. I just said that multicellularity basically emerging in a, as a way to shut stem cells or, you know, cells that are in an initial evolution of multicellularity, cells that are undergoing some evolution towards complexity, needing some kind of barrier from the environment, basically making themselves making, creating a Markov bank around them that makes it so they're not constantly, you know, being perturbed by the environment.
And that's the idea of how multicellularity evolves in the first place is to insulate niches, which will become stem cell niches from the environment so that cells that are undergoing, you know, very, very dynamic changes in short amount of time that therefore need to be kind of

sensitive to the environment, cannot be, need to be isolated from influence in the environment that would perturb this careful inference of cell type. Thanks, Dean. Yeah, so in this, in this figure too, you can see that there's, we're talking about ranges because we've got values on the horizontal and the vertical sides of the box. So, so one of the things that I think is really interesting is that once you, so you, we've all spelled it out, if we're dealing with active inference, we want to avoid something surprising, right? The whole, the whole variational aspect of this. So again, if we're talking about going from something that is, has low beliefs to something that gains in, I'll say gains in sensitivity without hopefully killing what I want to say next, if, if it's about signal adaptation to these top down conditions, which influence the final form taken, is there some way based on what you researched with this paper? Is there some way to know as you're approaching that place of energy exhaustion? Like, is there something that the cell tells you that it's, it's finding this too surprising? Because one of the things we, again, we talked about was that rate of change being so rapid, say, like the UN report on global warming, where we simply don't have the machinery to be able to adapt. Is there something that you found out through this paper that kind of said to you, as you were looking at these things, oh, wait a second here, we're just going to, we're going to, we're not, our, our intent is not to kill this thing off, but we're approaching that place where the environment is just too changing too fast. Yeah, I'm trying to give you a more to answer than hand-waviness, but let's start with the general answer first. So, yes, you do, it's, you can, of course, you can, like, one way you can quickly see it is in that kind of, like, something that you get with the prediction error plot, that you also get with the free energy plot, if you, right, in a perfect world, you'll get, like, a nice smooth, like, you know, kind of exponential decay of the free energy function. If that keeps changing, bouncing up and down, that will be very problematic. I did some simulations afterwards, we're not included in this paper, but based on exactly the same stuff, where I was including a time, based on top of this time sensitivity, you know, increase, that I use for basically, you know, increasing the sensitivity over the time, but having a low in the beginning. On top of that, I put in a pulsing, like a sinusoidal or a rectangular function, which was, we were just trying to simulate what I'm doing in my experiments in a lab, where basically, you know, I pulled certain signaling inputs to my organisms. I wanted to see, basically, if I randomize that, or if I made that in a certain pattern, how well would the simulation still be able to cope with that? And in the case where I had, like, a nice regular pattern of the, you're in the free energy land, and the free energy decay, you hardly saw that, you know, initially, initially, when that sensitivity was very high, it was, you know, you can see it there, there was like a big jumpiness and fuzziness. You could see it also in the belief updates, you know, basically, they were, they kind of emerged, kind of like, they diverged initially very strongly, and diverged in terms of their beliefs that they had about themselves. But then kind of afterwards, when the sensitivity overall was kind of acclimating, but was kind of varying around this acclimation point, you didn't see that much anymore in the, in the, in the, in the regular pattern, but in the ones were less regular, that was very much perturbed. And the, and the free energy also didn't actually quite, did not only did it not come to kind of like, to this asymptotic behavior, but actually started increasing again, basically,

just was really perturbed, and you didn't see that. So to answer the question, like, how do you see if, if you are not adapting correctly, even though you don't actually know yet what the, what your external state is, I would answer it's, how much are you switching back and forth between your beliefs? And I guess more precisely also like, how certain are you of that, right? This is something that we talk about in metacognition, which I've recently, from my last paper, I've been getting into and reading about metacognition and rats and primates and other animals. And the first step to metacognition is, as far as I understand it, is having a certain confidence about your results. They did this with rats, where they basically gave rats the option and this kind of cognitive task, that they could also just not answer the task, like not do, not push a lever, because I think it was levers, I forgot what the task was, but it was basically levers. And they had an option to not push a lever, for which they would get a lower reward than if they, a much lower reward than if they got it correctly, but more than if they got it wrongly. And so what they saw is that they actually would, in certain cases, when they hadn't learned it, when they hadn't learned the task correctly, they would just not push the lever.
And that is used as like a first instance of the metacognition, in a sense of they must have been unsure about what you do, and then chose not to do it. So I think that's kind of like, if you, it's one thing if you kind of lean back and forth, because you always think, well, you know, I'm this and that being unsure is normal, but kind of, kind of constantly going back and forth between certain states of sureness is also problematic. So that's the generalized answer to this.
Yeah, I think the interesting thing there is the emphasis on the back and forth, metaphorically speaking, we tend to give a lot of attention over to the balance piece. But what you're talking about, when you're talking about the environment is not the plank as much as a moving fulcrum. So how do you adjust? Right? The balance, I guess, is the outcome, but it's the process itself is the back and forth.
So yeah, I wondered about that, because some things you do want, from an epigenetic standpoint, you do want certain cells to die, you want their energy to expire.
Absolutely. I mean, I'm...
Right? That's part of the back and forth, as opposed to the balance, you know, there's no balance metaphor you want there, you want them to die. And they're not dying. So that's really interesting. Thank you.
Yeah. And I wanted to add one more kind of attempt to answer this more specifically, is there's a paper by, I'm probably pronouncing his name wrong, Joffoli, which was, if you just Google valence and active inference, they basically looked at, you know, valence in terms of like, judging if your actions are, you know, certain like, are they better or worse, basically, is there, are you assigning a sign, positive or negative to outcomes? And they were doing that in active inference schemes, and they were defining it. And, you know, there's other definitions, not everyone's going to agree about this, but they were looking at it in an active inference scheme about the rate of change. They were looking at the first and second order derivatives of how... I'm saying this correctly... about how the prior beliefs were updated to the precision, like how important, like how sure they were about what cell type they were. And so basically, in the second order derivative, basically, how much would they change at any given time in their beliefs, that's what they... where they plugged in valence into different mode that gets more complex than that,

but that's the gist of it. So that, I think, is a... in the... within the framework of active
inference is a more specific answer to your question, is look at the rate of change by which your beliefs are
updating. And that, with respect to your uncertainty, of course, this... this whole... Karl has written about
this as well. I'm very interested in as well from the biology standpoint of view, is that stress... stress
sensing mechanisms, like stress is a universal driver. In biology, it's actually much more important than
rewards. I kind of find that we always talk about rewards, but biology is much more focused on stress,
because it's more informative and it's more important to deal with than... than reward maximizing. But of
course, you don't always say it's just negative, but... the... the point there being is... stress as in... as in terms of
for uncertainty, like not having... not minimizing your energy effectively and stably, that will lead to stress. And there's paper that
that's in humans. I think I mentioned last time as well, where they do experiments with humans and mice as
well, where they... very ethical, where they shock... they give electric shocks... I think they did it both
for mice and humans. And then they showed that the... so they were more stressed out, not just by this... like the
shocks themselves, but they couldn't predict... Right. The electric shocks. So it's that uncertainty of what
coming next. I think that is the most... if I had to like say one thing that is most indicative of... of
you going in a bad place. So if you are stressed because you have no idea what's happening and you
cannot even... if you know something bad's going to happen, you know when it's happening, that's a lot
less bad than if you don't know at all. One last really dark example to really make it to a dark place is
there is... I am from... I was born in East Germany. I guess I was only one here open the wall fell, so... but
we... you know, my parents were very much and there was this... now it's a museum, but it was a prison
for the... the basic political detainees in East Germany and East Berlin. And we went there as a class once
and they showed us all the things... all the techniques they used to interrogate prisoners, including
torture. And one of the torture devices was I believe in a Japanese device where you would...
it's retired. You... you put your head down on this device and there's a water that's dropping from
the top onto your neck. So you think at this point like that doesn't sound too bad, right? The problem
was is that the drops, you know, there's... there's a lot of viscosity in there. So, you know, a lot of
kind of fluctuations so the drops wouldn't always come at the same time and it kept repeatedly happening.
all time and time again. So what the... and the prisoners actually... people that used to be in prison,
they were actually the ones giving the tour. So, they were telling us, yeah, if you're on the device
after a couple hours or so, these drops will feel like hammers on your... on your neck. And apparently
a big component of that was the... not... the body not being able to adapt to it because, you know, the
drops don't come at a perfect five-second interval. It happens kind of at semi-fast,
randomised intervals. So, that... I think there's a huge part in... and... and not just looking at
stress over time but how you can adapt to that stress and... in order to adapt to stress. Adaptation
to me always involves a component of prediction. If you can't predict the source of the stress when it
happens, it makes things a lot worse. This has been shown extensively in humans and also in... in... in...

lower organisms. And I believe very strongly that there's something fundamental to any biological system to fundamental drive. If you... if you... if you... if you... if you... if you... prescribed... if you described to the point of view that... minimizing uncertainty in your environment is a fundamental drive of life, then with that you prescribe that minimizing stress and the source of stress over time and being able to predict it is also fundamental and lack thereof makes a very dangerous system.

Blue? Thanks.

So... all that's super interesting. Just... okay. So let me... I'm going to start at the front and work backwards. So in terms of uncertainty and... how that's... causes stress... I'm really kind of... I don't know... reluctant to use the word stress in a biological system. Like... I think about stressing a biological system as like pushing it out of equilibrium. But... but... maybe that's just a stimulus, right? Like I want you to rebalance and go to like a new equilibrium. Like... or... or like... turn on heat shock protein. Or... you know... there's a variety of different things. Like... many ways you can view stress as a stimulus to do something new or to perturb the system in a new way. But... but in terms of uncertainty, I think that there's a lot of truth in... how uncomfortable people are with uncertainty like at a cognitive level. And you can even see it like in the stock market before like the election is in. Before the election results, like the... the stock market goes crazy. And... because nobody knows. I mean... it's not that one candidate is preferable over the other. But it's just the unknowing is very... like... people freak out. So... so... and that speaks... I think to metacognition also. And then... so... I wanted to back up a little bit if... if I can... to your discussion of information flow and how that might be difficult to quantify. And Daniel, I don't know... can you flip to the section for me please on... the generalized flow section? I think we started to get into that a little bit last week. And... we did discuss it in... in the dot zero. But I have... like... I know that this is very... mathematically technical. And so... I just have a little bit of... of questions about... this. And maybe... you can help explain it in a way that's perhaps less technical. But... specifically... information flow... and Chris Fields is coming... to discuss the FUP for generic quantum systems next week. And so... I can... I can similarly interrogate him about... about some of... of this stuff. And I'm looking forward to it. But... you know... we... we have a... a discussion... there's a discussion in the paper about the probability flow... and... information... like a... a... probability current... I think is actually what is used. And getting into... information currents... like if you look up what is an information current... I think... there's like the idea of the von Neumann... information current... and... that like is explicitly related to quantum systems. So... is there any similar... like... do we have a way to measure information current? Because even when Mike came to the live stream and talked about... expanding cognition... biological cognition... computational boundary of itself... there it was.

Sorry. But when you expand biological cognition... like there's... you know... you're basically expanding your... informational awareness. And so... is there anything like an information current... or... how... can you best...

relate that within a biological system? Is it like an expansion of... of... your computational boundary? Or... does it look like... just more... information... goes in... and comes out?

I think it's... depends on... depends on... on... what level you're looking at. I would say it's both. So... the... the... the... the second part... you mentioned... right...

the information... the information... comes in... and out. That's something we just have more access to... biologically speaking... experimentally... right?

um... and you can absolutely... you know... you can... you can look at... you know... you have all these... uh... sensors... for... for... for... for... for... for... for... for... for... for... for... um... genetic... um... transcription...

in response to that. So... so... so that's something that we can... you have the best chance of quantifying... experimentally...

and... and... in terms of... computationally... is something you can monitor... and then... also... couple that... with... what... Chris... um... good thing you mentioned... that... because... and I can... relate to him...

that's basically what... what he's also very interested in... right? Like... how... how... how... does physical energy... of course... played all... all... all... into this... in... in... in... in... in... in... in... metabolism... right?

now... this comes for free... that's something that... biologists... often... kind of... um... not biologists... actually... no... theoretical...

is... is

because there's people locally competing with you.

The same is true, right?

I mean, this is always like when you talk to,

when you talk to people that have problems,

the idea of why does entropy,

I don't wanna say names,

but that saying that entropy is always increase

if KRs in the universe increases,

how do we see much more structure evolving?

And then of course the answer to that is that,

well, just like in a bath of water,

if you have certain order outputs are gonna come together

because overall that makes,

that gives more degrees of freedom for all the water molecules.

So the entire system entropy did increase,

but you managed to find a solution that does increase that

by locally creating more order.

That's the drive of,

or I think of life overall.

The quote that probably,
the talk that got me into biology is,
was from a biophysicist in Heidelberg,
was saying life is not about energy,
it's for entropy.
But the two come together, right?
Yes, it's about how much information is available,
but then by extent of that,
once you are subscribed to that,
once you have to form these localized structures,
then energy becomes important again.
So, and that's where I think is needed right now,
when you have these highly spatially organized systems,
energy suddenly does become something that is valuable again.
So, yeah, I think I kind of lost my train of thought there.
The, the boundary of self, that's right.
The other information flow that was part of information flow,
we can measure that.
We have good ideas, what's lacking is,
and a quantum understanding of,
of, of how metabolism is really hooked into the information flow.
Whether or not it's built into the model,
whether it's ad hoc, just down the line,
you give it certain resources,
and after a while it just exploits them.
The boundary of self is interesting,
because, and of course in the cell,
we think, well, we have a, you know, there's a membrane,
the boundary of the self is fairly, fairly obvious.
It's not actually as easy as that, I think.
And of course, if you look at multicellular organisms,
it becomes, you know, tissue scale,
then that becomes a lot more difficult.
But even on a cellular level,
if you have certain receptors, you know,
you have certain activations of it,
then a model, any generative model,
not just in the active inference scheme,

but especially in the active inference scheme,
you also have a model of yourself, right?
And if, at what point your own,
your own, what you're secreting on the owner,
your signals that you secrete
also feed back into what you're sensing after all.
So it is kind of thought that that whole system
is very much perturbing cancer cells as well,
where the, how much, like, is it,
is, are you actually receiving things from the environment?
Are you basically just secreting and secreting
and you don't really react to it anymore?
So the, the boundary of self, what Mike,
if Mike already talked about this,
he will be, have been much more,
I can be about this,
but the idea is the boundary of self is encoded
specifically with time and spatial constraints.
So like how much your boundary of yourself,
what your, your self model,
be more general about this,
your self model depends very much on your sensual,
on your sensory memory.
Like how, how far do you sense and how,
and spatially, how much around you can sense,
and how much back do you have encoded as memory?
How much are effects in the past persisting?
So that, you know, so even though if you have a cell,
that's a certain membrane,
if, if that cell does not actually keep any track,
if that cell is not somehow sensing
or beyond a certain boundary,
and then it creates on boundary
by these highly peripheral cells
that are basically surrounding themselves with each other,
and suddenly the environment becomes mute.
So the boundary of the self has especially expanded,
I would argue.

So I think that's kind of the, the, the thought train here.
I, I'm sure hope that was somewhere
I would answer your question.
Awesome. Blue?
So just, I was curious,
you mentioned that there was a talk
that got you hooked into biology.
Was it Eric Smith and his discussion of biology
and entropy, because that's also fabulous.
It was not, I'm real, just seeing that.
And I was, damn, that's really embarrassing that the guy,
I don't remember right now the name of the person
that got me into biologies.
It's, I'm speaking English right now,
it's all in German back then.
And always kind of my brain then has problems
switching between those two in content,
as well as language.
I'll come back to me probably within this talk.
Eric Smith does a wonderful job though,
of, you know, discussing biology in terms of entropy.
And so if you are not familiar with this work,
I would just highly recommend it.
Yeah.
That really reflects on our discussion.
What parts of the self model are important
to carry forward?
And then what are the affordances for direct
and indirect self modification?
So that's one piece.
Like you mentioned the auto-crine signaling,
and that's not even a contentious component.
It just is being modeled in a different way
with active inference,
rather than just a molecular event that's happening,
that like a cells secrete molecules
that they also possess receptors to measure.
It's just putting that into a framework

of reflexive self modeling and stigmacy
and niche modification here.
So that's a very, very important point.
But just one other point that I thought was really,
maybe useful to highlight was this sensitivity.
And just like many of the terms in active,
we're kind of seeing that there is a narrow
or a quantitative sense,
and then there's a broader sense.
Because in any dynamical systems model,
you might hear about sensitivity of the parameter,
which is how much changing that parameter
changes the outcome of the system.
And so that's a quantitative sense.
And then also it was brought up
that it's important to model these like second order
derivatives, like how fast are things changing
relative to expectations,
or how fast are they changing how they're changing.
And that is what the generalized flow is,
is all those higher derivatives.
So we can use that quantitative sense of sensitivity
to look at how parameter changes
in the generalized coordinates of motion matter.
But also we can perhaps with cognitive modeling,
take this discussion of sensitivity
in the way that people are usually meaning it.
Like sensitivity of what is seen
on someone's emotional state,
and then model that using cognitive parameters,
like valence and affect,
even though that's quite a disjoint use
from the parameter sensitivity discussion
on the generalized coordinates.
So it was an awesome discussion with like,
what is stress,
and how does that relate to the parametric,
sort of neutral perspective on what stress

or sensitivity might mean,
just the model descriptive,
versus some of these functional
or phenomenological even ways
that these terms come into play?
Blue?
So that's super interesting,
thinking about sensitivity and stress
in just in terms of cognition, right?
Like, so I know that like people
that have sensory processing disorders
get like super aggravated by something
like a bright light or like a windy day
and just how sensitivity does play into
maybe the flow of information.
Like, because, you know,
I'm suddenly sensitive to this
and I'm also sensitized
to every other thing in my environment.
Like someone's screaming over here
and it's windy and it's hot and it's bright.
And then like, I become just so overstimulated, right?
Like people have these processing things
and it's sensory,
is that something that like has momentum, right?
Like, so you have enhanced sensitivity
and does that just like then all of a sudden
you're just, you're so sensitive.
Like we hear people say stuff like that.
Like, stop being so sensitive.
Just in terms of cognitive processing,
I thought that that was just super interesting.
I think that's an excellent point.
I mean, that's something that shows exactly
how important it is for humans.
And again, I argue low levels as well
in the evolution to be able to model our sensitivity,
to change our sensitivity, right?

We would never be able to have conversations
in low environments
if we couldn't change consciously our,
and sometimes subconsciously as well,
our sensitivity parameters.
And I think that's something also
that then also feeds into metacognition, right?
You'll learn like, oh, well,
I'm not paying attention enough.
So something to pay more attention to,
you're trying to tune this in.
And I think that's the, in the sense of,
on the cell level, I think where that feeds in is the,
is by learning over time, right?
But how much are being,
how many receptors are being activated
and how does that feed in downstream?
Then there's this component
where essentially the cell can up its sensitivity
by expressing more receptors by,
if it's not understanding its inputs,
it can increase those proteins
that are responsible for sensing these things.
And there's also different,
if you talk about iron channels specifically,
and in the lab we work with that,
there are different versions of iron channels
that have different rate constants.
The same is true for other,
for active proteins as well, such as pumps.
So that becomes a lot more,
you know, explicitly modulating sensitivity over time
in a way to, you know, better react
and cope with, adapt to its environment.
Again, adaptation, keep in mind,
involves prediction in most cases.
If not explicitly, then implicitly.
Awesome.

And by the way,
the guy is Michael Hausmann in Heidelberg
before I turn around my grave.
Oh, that was the person coming to physics,
and to biology, biophysics.
Cool.
Just to kind of connect that change in sensitivity
to some of those mechanisms.
So if it's the sensitivity to a given hormone
or a given circulating molecule,
then that sometimes gets accomplished biologically
by changes in like the membrane receptor density
or changes in downstream signaling pathways.
So even if the amount of receptor
in the membrane stays constant,
it's like you can release more
of the neurotransmitter in the synapse,
or you could have more or less receptors,
or you could have more or less
of all these regulators
and phosphorylating proteins, et cetera,
in the downstream signaling pathways,
because there's a lot of complexity in that space,
but there's a lot of knobs,
including rates of change
and rates of rates of change
and lag effects with transcription factors,
and all this other,
basically anything is possible.
And then at the organism level,
just to give one,
and then anyone else maybe at some other level,
like it's kind of like a nest mate ant
being sensitive to interactions.
And then a nurse who's a younger ant
is like initially more susceptible
or more sensitive due to multiple reasons,
like at the antenna,

at the brain,
in just spatially where the nurse is.
And then there's this development
towards being a forager,
being sensitive to forager cues,
finding oneself in spaces
where there are forager cues
that are useful.
So there's a lot of ways
that these changes in the type
of sensitivity,
again, are already things
that we talk about
in developmental biology.
And so it's so cool
to see how that is connected
to the molecular mechanism,
the affective angle,
and then with the underpinning
of the generalized flow.
Yeah, Dean?
So you guys can push back
on this hard if you want,
but it's interesting.
I think one of the terms
that I coined
was that stress programming,
or at least programming
with stress assumed
was basically learning,
either formally or informally.
That's what it is
because there's a certain amount
of stress involved,
be it positive or negative.
But kind of through that viewfinder,
I think it's interesting
that back to the back and forth thing,

stress programming can be safer
or it can be riskier,
as long as it's not
sort of killing the learner.
It's safer in the sense
of as intervention.
I think this paper
showed that you can intervene,
but riskier is as distribution,
like sending something
out into the unknown.
It can be safer
as teaching or taught.
It can also be riskier
as unteaching or untaught,
sort of just sort of
throwing somebody
into a situation
and see whether they can paddle
or not.
The last thing is
it can also be seen
as both modeling
in the safer sense
because we've reduced
some of the variation,
or it can also be riskier
as in terms of coordinating,
meaning the person
has to kind of figure it out
as they're flying the plane.
So I wonder what you guys think.
Do you think that,
can we,
is there some agreement
that learning in general,
be it formalized
or informalized,

has an aspect
of stress programming
that kind of is assumed
or from my way out?
Awesome question.
Blue or Franz,
what do you think?
I don't think you're off.
There's definitely,
of course,
some of the explicit
incorporations
of stress into learning,
but your question,
I think,
is like,
is it,
is kind of always
implicitly there,
even if not explicitly?
So one interesting
kind of fact,
this,
because Blue mentioned earlier,
that stress can be interpreted
as like human perceived stress
hormones,
but can also just be like
perturbation
from equilibrium,
from homeostasis.
And they actually do
kind of,
this has been reached,
this has been combined
in homeostatic reinforcement learning
where they kind of implicitly,
explicitly make

the reward
and reward-based,
you know,
reinforcement learning
about the homeostatic
off point.
And then stress
becomes suddenly
a lot more explicit
in that sense
of perturbation
from your homeostatic
set point.
In the active inference
scheme,
stress is basically
just your,
you could just define it
basically as your,
you know,
your overall
difference.
And again,
like with those
bands I was talking
about,
like how much
is your,
is your,
is your private belief
certainty changing?
Are you minimizing
uncertainty
over time or not?
it is,
it is hard.
I think it's hard,
it's hard both

to completely
say stress
is not involved
and without
saying that,
well,
you're just calling
everything stress.
So it kind of,
it's really,
it depends on what,
how you define the stress.
But if you think
if you go,
go with something
like,
as in like stress
as you're not
right now
in an energy
minimized state,
you're not in a,
an active inference
scheme,
or you're not
in a,
in a homosex
set point
in a more basic
scheme,
then stress is,
I would argue,
ubiquitous in learning.
Yes,
because without that,
if you don't have
a difference to that,
if you don't actually,

if you don't have
any drive to learn
because you're not,
you know,
you think you're,
everything's perfect,
then it's hard
to,
to have learning.
So specifically,
if learning is about
with the environment
and context,
what is,
what are you getting
from your environment?
Are you trying to adapt
to it?
Are you trying to learning
in a sense of like adapt
to it and find a certain
equilibrium with it?
Then I think stress
becomes very much explicit
and it's ubiquitous
in learning.
Okay.
I just want to add one thing
and that's perfect
because I always talk
about the minimum of two
and I think you're,
I think you're absolutely
right, Franz.
I think you have to then
look at stress
as being either voluntary
or involuntary.

I think that's the minimum
of two that you have
to look at
because I think
if you have voluntary stress,
it's,
it's,
it's,
it's measurably different
than the involuntary type.
So,
yeah,
you're right.
I don't think it's,
I don't think we can just
throw under everything
under one umbrella
and make it a monolith.
I think we have to partition
it right from the very get-go
and say,
all programming,
be it a curriculum,
whatever,
whatever the curriculum is
that those stem cells have,
that's,
some of it's going
to be voluntary
and some of it's going
to be involuntary
based on the situation
that they're,
they're placed in.
So,
again,
I wasn't trying
to pull it in that direction

but I didn't bring up stress
but I really liked
that it was brought up.
So,
okay,
yeah,
blue,
blue.
Sorry,
so just
from a
neuroscience
perspective,
it's interesting,
like the voluntary
and involuntary
stress thing
is so different
and I feel like
maybe Daniel will remember
we discussed that
at some point,
like when you're exposed
to a stress
that like you do
to yourself on purpose,
like your strength training
or something like that
versus,
you know,
some,
some externally
imposed stress
that you didn't volunteer for.
I think that that has
a difference in affect.
I feel like we looked
at a paper,

I can't remember
off the top of my head though,
but in terms
of sensory processing,
it's really interesting
in neuroscience,
we actually adapt,
right?
So,
so you can listen
to like your,
you,
this is a great example,
like you turn
on the car radio
and like you're listening,
jamming out in the car
for,
I don't know,
a while,
20 minutes,
30 minutes
and then suddenly
it doesn't seem as loud
and so you turn it up
and then you're jamming out,
jamming out,
jamming out
and still,
it doesn't seem as loud
so you turn it up
because the same level
of stimulus
produces a decreasing sensation
and that's true
in terms of,
you know,
visual,

like when you first
turn on the light,
you're like,
oh,
it's so bright
but then you adapt
and so in,
in terms of sensory processing,
we have a mechanism
for adaptation
and I wonder
about actually cells,
like not neurons
because this is like,
I'm specifically talking
about like a sensory processing way
but I wonder if cells
do the same thing,
I mean,
I guess like,
you know,
the receptor shuts
or the receptor's triggered
and then it can't get activated again
for some certain amount of time
so even just receptor binding
in and of itself
is like an adaptation mechanism
like you can't constantly
like dump the chemical
into the receptor,
open the channel
or whatever,
like that's not
like biochemically possible
but I wonder if,
you know,
cells have some other

adaptation mechanism
outside of this
like channel opening
or closing
or just this,
is there more than
like the mechanodynamics
of the channel
that enables
cellular adaptation?
I can think of a couple ones
that I know
we're an expert on this
but there's,
you can think of first
on the epigenetic level
if you basically just,
you know,
recruit more histones
basically have a more
tightly packed chromosome
you're going to have
less genetic expression
no matter what
kind of signaling input
you get to the nucleus.
That's one level
I can think of it.
Another one is
mechanically
so the right,
most of the,
I hope I'm not
overgeneralizing
but I think most of the,
you know,
signaling against
that receptors are sensing

and then has to be transported
to the nucleus
somehow to cause
transcription.
That is usually directed,
right?
That has to crack
double-cross cytoskeleton
so if you change
the mechanical properties
of the cytoskeleton itself
and,
again,
yeah,
I'm out of my league
but that's something
I could very well imagine
to ask someone
that's smart on me
on that topic.
lastly,
it's,
if you talk in the intercell,
intercellular context
then,
you know,
things like,
you know,
what we,
in neuronal context
you have these,
you know,
the synapses
in the non-neuronal context
you still have gap junctions
and other complexes
that basically allow
transport between cells

and those can also
be modulated
by the cell
so then there
you can also get
habituation
and definitely
know of examples
where people
have shown habituation
and even other
there's active research
on showing
that there's more
habituation
but also actually
the learning
of a certain pattern
as well.
I think that's true.
So I'm really interested
in the cytoskeleton
because a lot of,
you know,
I mean,
the receptors
are in the membrane
embedded in the membrane
but we don't really,
there's not a lot of work
that looks at
like the cytoskeletal dynamics
with respect to
like modulation
of receptor dynamics,
right?
So like that's a field
that I think

or I haven't seen
very much work
in terms of
how does the underlying
cytoskeleton
contribute to the properties
of the membrane
and how a cell
might be behaving
and just in terms of like,
you know,
things having to get
into the nucleus,
I don't know.
I'm iffy on that one.
I think,
and it's just
because I did
very early work
like transfecting cells
in a variety
of different ways
and, you know,
to get a gene
into the cell
to get the cell
to express the gene,
you have to get it
through the nucleus,
right?
How does it get
into the nucleus?
Like nobody can answer you.
So like if it doesn't
have a nuclear target on it,
like how is it going?
So is it just like
osmosing itself through

or like just
how is that
exactly happening?
And so you call the vendor
like,
oh,
I'm trying to use your product
whether you're
electroporating
or, you know,
using some kind
of lipid mediated
transduction,
they don't know.
They have no idea actually.
So it's just
a super interesting
prospect.
And I think that
in terms of both that
and the cytoskeleton,
there's a lot of like
dynamic remodeling
that occurs
during cell cycles,
right?
Like during the phases,
you know,
like when a cell is dividing
and growing
and becoming two cells,
there's,
and so like the nuclear
membrane
point,
but in terms of terminally
differentiated cells
like a neuron

or something
that's not undergoing,
you know,
constant change
and remodeling,
it's curious to see
how that can happen.
And I'm interested
if anybody has any,
any good like
hardcore mechanistic papers
that you want to point
out to me
or if anybody listening
in the live stream
wants to drop them
in the YouTube chat,
I would really love
to take a look at them.
Bill.
I really like Dean
returning to this
idea that
there's the,
there's a continuum
or there's multiple
archetype dimensions.
So not that it's all
going to map on to just two,
but we know it's
at least locally
going to be at minimum two
that there's the kind of
one mode of
interacting
in an educational
or training setting
with a teaching

and a taught
that could still be
in an interactionist
or an instructionist
way
versus the unteaching
and the untaught.
So that's
the difference
between the athlete
having their form
like observed
and a type feedback
versus when they're
on a sojourn
away from that
sort of a training
context.
And so there's like
an environment
or there's a setting
where there's
some kind of
stress as fundamental
and then
there's another
type of environment
where the different
kind of stress
is fundamental
and then
Franz,
it was an awesome
point
moving from that
basic homeostatic
framing
where stress

is usually
considered at the
first derivative
or the state.
So it's like
for a thermo
organism,
thermal stress
is going to be
when it's hot
or cold
and then that's
going to be
like a bowl
or it's a V
or it's a bathtub
or it's some other
thermal stress curve.
It's about temperature.
Moving into the
generalized coordinates
of motion
with temperature
and change of
temperature
and change of
change of temperature
so all the higher
derivatives of temperature
that expands
the space a lot
because maybe it's
possible that a
temperature that you
slowly reach
like over
10 years
is a different

amount of stress
than just jumping
there instantly.
Time matters
and so the rates
of change
captures that
time dependence
in an instance
with a snapshot
vector.
So that's sort
of the formal
use of how
we can
look beyond
just the stress
on temperature
and then
we introduce
the whole
cognitive stress
because you said
well for the
active inference
entity
this is sort
of like the
first
resonance
and this is
the one
whose
violation
of integrity
results in
physical death
but then

stress
moves to
higher
or different
analogous
settings
and that
could be
how fast
am I
learning
and as
pointed out
by Dave
in the chat
and by Dean
and others
like it really
matters whether
somebody has
agency
or affordances
to change it
and so my
closing point
there with
the voluntary
involuntary
is like
grad school
and stress
because
or any
research
or any
educational
environment
but one

that many
people experience
is like
it can be
stressful
it's also
a very
often
privileged
setting to
be in
and one
is taken
care of
sometimes
in a way
that even
like neighbors
won't be
and so that
is something
that like
I saw
first and
second hand
and just
thinking about
the stress
it's like
why is it
stressful
to just
read a few
papers
some days
or something
like that
so just

kind of
interesting
with what
is voluntary
and involuntary
how we
commit ourselves
to different
kinds of
stress
of what
kind
it's like
well I
wanted it
to be
novel
but not
that
novel
and how
we actually
make those
action
selections
in
uncertain
environments
blue
so just
to add
on to
the
voluntary
versus
involuntary
like grad
school is

an entirely
voluntary
endeavor
obviously
but like
high school
is not
and like
I dropped
out of high
school at
16
like as soon
as I was
old enough
to drop
out I was
like I'm
out
like I'm
not doing
this compulsory
regimented
program
like sorry
no regrets
right
like I started
college
obviously
have a PhD
like did
well
but it's
the difference
versus
involuntary
versus

voluntary
like when
it's
compulsory
it becomes
torture
like
I mean
just my
two sets
we had
an interesting
discussion
about
computation
group
recently
about
you know
this whole
idea of
how do you
quantify
any kind
of agency
and
Josh
in that
context
brought up
this idea
instead of
talking about
agency
which is
hard to
quantify
you can

talk
about
empowerment
and
in the
computational
sense
empowerment
is about
how much
do your
actions
how much
can
your
actions
influence
your
sensations
again this
makes
very clear
an active
influence
scheme
right where
you have
this
market
so
to
build an
example
right
if you
if you
don't
if there's

no actions
you can
take
to change
it
that's
you know
we can
think about
it as
natively
stressful
but we
can also
define that
fairly
clearly
in a
computational
sense
using
something
like
empowerment
so
it's
and that
again I like
that you brought
this up
because this
drives home
the point
that a lot
of these
fundamental
things we
understand

about stress
in human
context
anyone that's
ever dealt
with depression
or people
that suffer
from depression
knows that
it's never
really about
the
the
the
objective
total amount
of stress
but how it's
being perceived
by that person
and I think
this empowerment
helps a lot
this definition
empowerment
helps a lot
to understand
that better
how much
control
you have
over that
there's other
things that
of course
we enter
that

but it also
makes
then fairly
clearly
that the
same
a lot
of the
fundamental
principles
must hold
true
in lower
organisms
as well
because
these definitions
are of course
you know
as soon as
you prescribe
actions
of course
you can argue
that there's
no real
actions
in cell
biology
then that's
okay
that's just
then don't
want to talk
about that
level
but if you
do look

at active
states
in a
cellular
context
then you
do get
then you
can fairly
simply
in an
active
inference
scheme
but other
ones as
well
define
empowerment
measure
empowerment
over time
by looking
at how
actions
actually
change
the
sensory
states
of the
cell
and then
definitions
of stress
from that
point of view
also become

fairly
explicit
can i
ask you a
question
yeah i can
ask you a
question because
you mentioned
in the when
you were with
us in the
point one
for a bit
there and i
don't know if
you mentioned
it today
you said
you went
to mark
i think
and you
said i want
to do
something
cool
we're going
to bring
this back
to the
voluntary
and involuntary
piece
and then
he
and then
i think

he said
i think
you said
i'd never
heard of
active
inference
before
i was
pointed in
that
direction
to carl's
lab
so
in the
in the
in the
context
of what's
voluntary
and involuntary
when you
when you
when you
said i want
to do
something
cool
that sounds
like you
wanted to
do
something
voluntarily
then you
got
then you

got this
deep dive
into
active
you
pushed
that
up
against
your
biology
background
which is
something
that
you've
obviously
done
because
you
like
it
right
and you
wanted
to
finish
off
so
so
going
forward
now
that
that
voluntary
piece
that got

you
into
the
active
inference
what
do
you
see
the
active
inference
doing
in terms
of
this
empowerment
question
because
Daniel's
written
out
here
with
a
question
mark
behind
it
so
how
do
you
feel

more
I'm
just

curious
what
you
think
now
in
terms
of
what
sort
of
empowerment
you've
sort
of
become
in
the sense
that
now that
you've
got
this
sense
of
what
active
inference
can
do
and
what
some
of
these
math
formalisms
provide

and
what
the
statistical
world
bridges
what
do
you
think
in
going
forward
in
terms
of
the
reapplication
of
what
you
what
you
pulled
out
of
a
different
bucket
and
then
set
down
in
the
bucket
that
you're

most
familiar
with
so
you
mean
like
what
I
pull
that
on
a
personal
level
like
for
my
life
decisions
yeah
like
when
you're
going
to
be
going
forward
and
trying
to
use
that
active
inference
sense
in

terms
of
some
of
those
decisions
and
what
kind
of
environments
do
you
think
you
see
yourself
projecting
yourself
into
that
you
may
not
have
projected
yourself
into
prior
to
sort
of
engaging
or
encountering
or
exposing
yourself

to
active
inference
inference
that's
a
question
I
think
in
the
end
it

it
it
my
wife
and
I
we
both
love
this
word
serendipity
which is
the
concept of
happy
accidents
which
there's
all kinds
of
connotations
but
in

active
inference
scheme
or
any
you
know
when
you
look
at
what
is
the
information
from
you
what
the
idea
is
that
you
if
you
have
tumor
if
you
have
so
it's
going
to
!
very
strong
beliefs

then
whatever
you
if
you
are
confining
yourself
and
you
don't
allow
variances
to
change
your
mind
then
you're
almost
always
going
to
be
upset
because
you
never
you
know
your
precision
is
too
high
right
there's
a

paper
going to
be
coming
out
soon
about
this
collaborator
as
well
looks
about
precision
in
this
context
as
well
so
that's
something
that
any
active
inference
model
will
know
that
you
have
to
keep
that
in
mind
your

precision
for
me
personally
it
really
is
an
idea
of
like
I
think
every
there's
not
an
idea
by
myself
any
stretch
but
any
human
tells
stories
about
the
environment
but
most
about
themselves
and
the
storytelling
is

in
my
sense
nothing
else
than
doing
a
model
and
making
a
generative
model
and
setting
that
so
I
think
for
me
I
am
too
young
to
be
giving
life
advice
but
for
me
to
achieve
any
kind

of
happiness
you
have
to
be
careful
of
what
story
you
tell
about
yourself
and
how
strictly
you
can
find
that
on
things
and
there
is
definitely
advantage
to
yourself
I
think
personally
I've
always
had
better
luck

personally
I
think
by
not
letting
not
define
that
too
strictly
and
really
looking
into
what
your
environment
throws
at
you
and
I
think
that
helps
and
I
know
personally
people
that
have
a
very
hard
time
with

that
and
that
have
very
strong
modeled
so
if
you
know
I'm
going
to
get
married
at
that
age
and
do
this
and
that
and
then
when
things
don't
happen
that
way
I
think
it
leads
to love
and

happiness
so
in that
context
what can
I do
for you
I mean
I
think
it
makes
things
explicit
that
we
all
hear
about
some
level
and
psychology
things
just
people
giving
you
advice
parents
I
don't
want
to
say
you
need
any

of
those
insights
you
can
get
this
lots
of
ways
but
what
I
like
about
that
inference
scheme
it
makes
the
it
makes
the
biology
the
biggest
advantage
you
get
a
very
explicit
structure
about
information
flow
for

very
simple
systems
and
because
it's
been
primarily
applied
in
neurosciences
you get
for us
intuitive
results
from that
what happens
if you
have
try
precision
what
happens
if
your
energy
isn't
being
minimized
all
these
things
I
can
fairly
easily
explain
that

in
the
context
of
human
behavior
because
that's
been
applied
and
we
all
have
true
understanding
but
I
think
there's
for
the
biology
point
of
view
there's
a lot
of
things
we
can
learn
from
that
if
you
look

at
the
mathematical
definitions
then of
course
you deal
with the
baggage
that you
get
from
the
neuroscience
which
you have to deal
a lot
with
which is
fair
but for
your life
I think
it just
helps
you
once
you see
really
how
much
study
has
been
put
into
mis
inference

bad
inferences
and of
course
people
doing
tests
in
an
actual
inference
scheme
about
psychological
behavior
tasks
states
then
I think
that's
one lesson
I can
definitely
learn from
that
if I
hadn't
already
is that
be mindful
of
your
price
be mindful
of
the
rate
of

change
like
what are
you
stressed
out
about
right
now
what's
the
worst
uncertainty
coming
from
and
how
much
should
that
throw
you
off
right
and
this
is
in a
sense
to
bring
all
this
home
a
little
bit
the

biggest
limitation
of
any
algorithm
of
any
model
is
that
you
are
defining
the
model
beforehand
not
just
an
active
inference
scheme
but
any
there's
papers
that
come
out
that
criticize
the
idea
of
artificial
generalized
intelligence
because

you
always
have
predefined
model
and
life
on
that
sense
but
that
definition
does
not
do
that
life
always
in
biological
systems
evolution
generates
a bunch
of
diversity
in
its
background
and
from
that
they
pick
the
best
path

constantly
evolve
the
generative
model
so
that's
kind
of
limitation
of
any
other
algorithm
which
can
be
overcome
by
changing
different
algorithms
together
by evolving
systems
by having
models
make
models
there's
some
cool
work
on
that
too
but
that's

something
you
need
to
take
both
computationally
scientifically
but
personally
take
in
mind
of
at what
point
do
you
change
your
own
model
and
allow
for
flexibility
that's
what
makes
human
right
now
still
that's
one
of
the
soft

acts
that makes
humans
superior
I would
say
to any
kind of
algorithm
right now
is because
they all
come
predefined
with a lot
of the
hard
model
again
I'm
saying
this
now
I also
don't
actually
agree
with
that
100%
I don't
actually
believe
in
fundamental
differences
between
different

intelligences
but
that's
something
that
at least
intuitively
makes
sense
to
whether
or
not
that's
true
scientifically
I have
my
points
but
that's
going
off
the
rails
awesome
wow
a few
things
like we
often
talk
about
concordances
with
different
systems
and

it
brings
the
cognitive
apparatus
that was
developed
for
humans
and
starts
to
at least
instrumentally
if not
from a
realism
perspective
also
project
that
and so
it
allows
as a
trans
discipline
people
to
talk
about
systems
and
map
their
analogies
and
models

in a
way
that
would
help
us
find
the
resonances
between
cellular
metabolism
and
economics
kind of
taking a
vague
feeling
like those
might have
some
similarities
into
using
the same
notation
terms
forms
etc
so that's
one very
interesting
piece about
active
inference
and
also
the

limitations
in terms
of
at
what
order
and
how
much
time
and
attention
is
allocated
to
this
structural
aspect
and
how
many
different
families
of
structures
are
provided
and
then
to
your
well
communicated
uncertainty
about
if and
how
biology

is
different
from
other
kinds
of
processes
that's
an
awesome
question
especially
connecting
back to
your
point
about
the
reservoir
of
energy
versus
how
locally
there
can
be
a
different
energy
or
entropy
balance
so
it's
almost
like
there

might
be
little
pieces
that
can
be
mapped
to
digital
signal
processing
with
super
high
fidelity
and
that
will
be
taken
as
a
win
for
the
realists
and
then
there
might
be
other
parts
where
the
instrumentalism
bridges

because
we're
just
making
a
model
of
this
one
smaller
thing
and
this
one
larger
thing
and
then
in
cases
where
this
is
fully
computationally
defined
or
fully
described
then
there's
like
a
case
for
a
real
what

does
empowerment
look like
at
maybe
like
a
lower
organismal
level
or
even
like
a
cellular
level
how
like
is
cellular
behavior
or
lower
organism
behavior
like
voluntary
versus
compulsory
and
like
is
there
a
way
to
measure
that

like
obviously
like
you can
trap
an
animal
in
a
box
and
frequently
like
you know
will
try to
get out
but
if
you
don't
it
would
it
go
into
the
box
and
hang
out
there
like
does
that
happen
maybe
I don't

know
does
anybody
know
of
any
examples
of
like
what
empowerment
looks like
at
like
that's
just
it's
a super
interesting
concept
to me
in the
in the
sex
if you
take
that
definition
of
how
much
your
actions
change
your
sensory
perceptions
then

definitely
definitely
you
I would
if
once
you
have
a
definition
it's
of
course
built
on
what
your
actions
are
and
that
is
fairly
easy
because
you
don't
have
to
go
into
this
route
but
it's
voluntary
not
you

just
something
you
know
you
look at
the
flow
of
states
your
marker
blanket
which
you
can
show
forms
naturally
which
car
is
shown
and
then
from
that
point
of
you
once
you
have
active
states
then
you
can

measure
that
fairly
easily
so
as
example
I
would
argue
if
cells
actually
interact
with
each
other
and
really
change
other
cell
types
here
is
a great
example
morphogenesis
is traditionally
fairly
established
driven by
large
organizing
and
it's
always
a

subgroup
of
cells
that
will
drive
the
ones
I
would
argue
then
that
those
cells
are
probably
very
much
in
power
because
their
actions
cause
differentiation
of
cells
around
them
which
then
will
very
much
and
they
are

going
to
do
certain
expression
of
cell
types
well
so
then
that
will
they
have
action
very much
control
that
whereas
the
receiving
end
have
a
low
impact
because
their
own
signaling
probably
isn't
really
because
they
often
just

don't
have
that
they
don't
make
a lot
of
that
signaling
cell
that
it's
being
organizing
element
so
they
have
a
lower
let's
think
of
that
you
asking
me
what
anyone
yeah
what
as
our
guest
or
anyone
else

is
welcome
to
whether
to
go
I
personally
like
to
get
to
where
there
comes
more
than
kind
of
quality
of
like
making
actual
like
instead
of
just
like
making
these
models
of
active
inference
you
can
measure

on
the
hand
if
you
can
really
map
those
two
together
right
like
in
the
sense
that
I
mean
a lot
of
work
went
into
this
whole
SPM
software
that
they
developed
in
London
and
at
this
point
you

actually
can
get
fMRI
data
and
then
you
can
do
dynamic
causal
models
and
you
can
really
postulate
different
causal
models
and
do
a
Bayesian
belief
updating
you
can
do
hypothesis
that
you
see
if
we
get
to

that
level
in
a
morphogen
for
example
and
get
the
high
fidelity
images
of
a
developing
embryo
that
you
would
get
over
time
get
signals
that
you
can
measure
and
feedback
into
your
model
and
really
make
really

test

of

what

cause

a

model

like

what

structure

would

have

caused

these

sensations

that

you

that

you

receiving

that

would

be

a

huge

step

forward

and

microscopy

great

advance

on

that

one

example

I can

give

is

a

friend
of mine
works
on
that
Dimitri
Krum
and
he used
to be
in
Heidelberg
and he
was
light
sheet
microscopes
there's
a lot
of
other
groups
he's
a
master
student
PhD
student
so
I'll
know
about
that
from
him
but
there's
groups

that
do
high
fidelity
light
sheet
from
two
sides
really
nice
like
near
confocal
resolution
but
with
high
temporal
resolution
of
their
organism
doing
the entire
process
of
you know
from
castellation
further on
that's
you could
collect
where you
can really
start to
make

postulate
up
cause
in an
active
inference
framework
and then
they
basically
retest
where's
the
actual
palm
there
and you
can
retest
the idea
that we
already
have
as
mentioned
as I
mentioned
in
morphogenesis
and retest
out how
much
that fits
into
our
notions
of
update

in
actions
I
would
love
to
see
that
in
my
lifetime
wow
super
awesome
so
just
to
kind
of
restate
that
SPM
has
been
developed
in
the
capacity
of
behavioral
neuro
imaging
and
so
it
allows
a
combination

of
observable
features
such
as
what
button
somebody
pressed
or
what
image
somebody
was
shown
and
also
observable
features
of the
environment
like
the
measurements
coming
in
off
of
the
EEG
or
the
FMRI
or
the
MEG
and
it

allows
integrated
modeling
of those
different
kinds
of
parameters
along
with
the
dynamical
causal
modeling
on
the
underlying
dynamic
system
identification
and
there's
a lot
more
in
the
SPM
documentation
on
that
and
so
ACDIMPH
here
along
with
some
advances

in
for
example
the
ability
to
look
at
the
process
of
embryogenesis
and
track
cellular
location
for
a
whole
organism
developing
through
time
then
that
can
be
fit
as
if
those
are
the
fMRI
measurements
of
this
embryo

SPM
and
then
the
underlying

mechanics
of
development
like
a
hypothesized
morphogen
gradient
or
a
known
or
hypothesized
mechanical
relationship
that's
part of
the
model
that's
like
the
underlying
neural
systems
model
in
SPM
and
so
it's
kind

of
a
great
dream
but
if it
can be
stated
if
it's
likely
to
exist
sounds
like
you're
taking
some
actions
that
will
reduce
your
surprise
about
it
yeah
pretty
cool
so
on the
microscopy
note
I'm
like
getting
ready
to

receive
and
set
up
IO
AFM
I'm
really
looking
forward
to
like
the
multi
parametric
capability
of
like
I
can
look
structures
top
left
also
like
do
fluorescence
imaging
so
I'm
really
excited
to
have
the
affordances
available

on
this
same
like
the
micro
advances
are
great
like
I've
used
an
AFM
book
and
you
know
it
gives
you
the
biomechanics
of
sample
but
not
not
it's
I've
never
also
been
able
to
look
at
the

sample
you
just
the
mapping
without
being
able
to
see
it
so
I'm
really
excited
to
be
able
to
see
and
measure
what's
going
on
in
a
cell
that's
the
skeleton
parameters
that was
discussing
earlier
about
how
that

you
know
manipulates
or enables
a cell
to do
different
things
cool
cool
let's
look
at
any
other
things
or
also
we
can
sort
of
give
a
closing
round
whatever
people
feel
like
so
one
thing
I'd
like
to
find
really

about
the
aberrant
signaling
figure
so
I
don't
think
we
went
into
figure
five
very
much
last
time
in
the
dot
one
and
maybe
you
could
talk
about
the
aberrant
signaling
rescue
kind of
unpack that
for us
in your
own words
we're not

seeing the
slides
anymore
also
Daniel
I don't
know if
you know
that
or I'm
not
yeah
it's
also
kind
of
lagged
on
YouTube
so
I'm
recording
it
I'll
reupload
like
the
high
quality
one
it's
kind
of
like
weird
it's
a
normal

video
chat
but
the
YouTube
stream
link
is
like
disrupted
but
c'est la vie
yeah
so
actually
if you
don't
see
the
I'm
showing
figure
five
even
though
you
can't
see
it
so
just
oh
yeah
or
I
see
okay
we

can
do
that
too
yes
that's
good
great
and
thank
you
awesome
yes
so
the
so
this
is
the
the
top
release
the
non
stimulation
this
reference
right
you start
off
with
these
eight
cells
in
the
beginning
and

kind
of
have
this
vague
very
specified
low
prior
belief
of
what
type
they
are
and
as
they
move
out
that
intensity
changes
so
the
hue
tone
here
means
that
they're
now
more
the
more
sure
they
are

of
being
this
head
type
cell
or
in
case
the
B
where
one
cell
which
you can
see
!
have
been
out
that
had
reduced
sensitivity
to
movement
264
basically
had a
really hard
time
basically
first of all
it was
much
longer
unspecified

than its
neighbors
were
and in
the end
it achieves
actually
it gets
wrong
cell type
race
not only is it
at a wrong
position
there's
no
cell
in
the
original
one
at that
place
and
even
were
you
expected
to be
either
yellow
red
or
somewhere
in
between
but not
blue

as it
is
down
here
so
the
sensation
of
a cell
of a
cancer
is
I
term
this
that
way
because
I
think
that's
basically
the
first
step
in
cancer
initiation
is
mis
identification
of your
environment
and
your
place
in
it

right
the
original
paper
from
Carl
was
called
knowing
one's
place
the
cell
definitely
does not
know
its
place
so
this
goes
in
and
then
here
is
where
you
have
basically
the
same
as
above
but
the
overall
not

just
the
absolute
flow
of
actually
signaling
to
one's
cells
is
to
get
to
that
has
been
upregulated
so
that
each
cell
basically
now
drives
more
concentrations
of
cells
to
that
cell
and
basically
kind
of
tries
to

compensate
for
that
that
all
the
sensitivity
and what
happens
that
I
expect
at all
is
that
you
have
um
if a
cell
contracting
with it
more
here
so
you
don't
see
that
so
much
yet
you

see
this
cell
that

kind
of
initially
would
normally
gone
down
more
cell
push
out
in
the
end
here
what
happened
the
words
is
that
were
put
into
most
of
signaling
concentration
and
allowed
arrangement
and
allowed
this
cell
to
retake
a

place
that
is
conform
what
you
see
in
the
target
morphology
and
what
I
always
when I
talk about
this
figure
and
both
figures
really
is
you
do
not
in
the
simulation
at
any
point
in
time
change
target
morphology

all
the
actual
encoding
was
exactly
the
same
so
what
they're
expecting
and
where
they're
expecting
is
exactly
the
same
and
that
I
think
is
where
I
see
the
fundamental
promise
of
this
work
that
you
don't
always

want
to
change
the
code
we
know
that
we
have
higher
value
we
ever
had
and
yet
I
think
anyone
really
thinks
that
this
is
the
way
forward
in
all
cases
and
if
you
can
do
it
and

you
have
no

other
options
in
extreme
cases
that
is
absolutely
unwarranted
I'm
not
trying
to
dissuade
anyone
from
advances
in
genotherapy
but
I
think
the
genome
is
so
complex
so many
things
can
go
wrong
we
don't

know
all
the
details
that
if
you
did
not
have
to
start
messing
up
people
loading
I
think
that
would
be
preferred
and
also
from
a
capability
setting
you
don't
always
have
the
capability
with
high
fidelity
that

we
I
take
at the
core
of
my
experimental
approach
to
things
is that
I
want
to
understand
how
cells
interact
with
their
environment
and
work
on
that
manipulation
and
drive
them
towards
different
states
again
with
that
Wellington
landscape

that
we
talked
last
time
you
want
to
direct
the
floor
I
took
my
shirt
on
I
can
see
that
right
now
I
do
have
the
landscape
there
on
my
shirt
this
time
I
did
that
so
that's

like
what
are you
trying
to
redirect
the
flow
of
decision
making
not
try
to
manipulate
the
actual
presence
themselves
so
what
was
the
perturbed
signal
response
oh
sorry
there
yeah
the
perturbed
signal
response
so
what
was
what

was
yeah
how
did
you
perturb
the
like
did
you
cut
it
off
did
it
get
less
information
or
did
it
like
was
it
supposed
to
infer
an
incorrect
place
like
how
exactly
in that
one
cell
did
you

manipulate
the
parameters
of
that
one
cell
so
that
it
went
to
a
different
place
yeah
basically
let me
see
I
think
I
was
!
it
was
basically
done
in
the
decrease
in
sensitivity
is
that
what
it
was

I
try
and
make
this
specific
but
hand
waving
that
so
it
is
in
the
gradient
that
we
also
formally
induced
right
so
that
that
is
basically
the
how
much
of
a
right
this
the
how
much
actions

are
being
updated
based
on
sensory
inputs
and
we
can
put
a
gradient
on
there
and
the
sign
on
that
that
that
is
the
part
that
I
manipulate
here
so
but
that
basically
means
is
that
by
altering

the
response
capability
of
the
cell
how
much
how
much
sensory
update
how
many
how
much
to
what
extent
sessions
are
being
updated
based
on
external
information
flow
and
with
what
kind
of
you
know
power
on
top

of
that
that
by
that
you
basically
upping
or
reducing
the
sensitivity
to
the
environment
you
basically
you
misarrange
you
misregulating
the
again
the
power
essentially
but
that
sounds
stream
but
you
the
potential
of the
cell
to
act

with
environments
and
do
anything
about
and also
even
upstream
how much
sensing
of that
so in
this case
specifically
it just
wasn't really
sensing as
much of
its
environment
as it
was
supposed
to be
sense
it
it
almost
is
equivalent
to
the
argument
or the
dissolution
of sort
of nature

and nurture
partitioning
by just
taking a
complexity
stance
kind of
like
Evelyn
Fox
Keller
in
her book
the
mirage
of a
space
between
nature
and
nurture
you were
talking about
a target
morphology
but then
about how
changes in
the signal
we're about
to do
number 40
on the
quantum
FEP
paper
and
blue

and our
friend
Jason and
I have been
preparing a lot
for it
and I think
separating
quantum
from
the
electrons
and their
behavior
and just
asking
well what
is
quantum
strategy
or what
does this
quantum
statistics
or quantum
information
projected onto
biology mean
not how
it's been
approached
from the
mechanistic
question
which is
quantum
biology
well that's

about the
synapses
and their
quantum
mechanical
properties
or about
photon
tunneling
or proton
tunneling
and like
quantum
effects
but what
about just
using the
statistics
instrumentally
of quantum
and talking
about situations
with complex
patterns of
uncertainty
and observation
bias and
memory and
modeling and
all of that
and this
was like a
dot two
that kind
of like
opened up
a whole new
area for us

in our
discussions
with morphology
and like the
spatial and
the embedded
and then
cracked a
window
into just
another area
we'll go
into and
you brought
so many
awesome
insights so
we really
appreciate it
yeah that
talk should
be a real
pleasure
I highly
recommend
doing it
Chris Fields
was great
to listen
to and
it's some
exciting work
and if
you also
if you have
seen I'm
sure you
have seen

that the
particular
paper from
Carl Frist
he has a
section
there on
quantum
mechanics
and it's
a very
cool section
because he
actually derives
I think the
Schroding
equation
from
essentially
from my
point of
view
from just
a definition
of what
your log
probability
density
is
very cool
kind of
you know
I don't
know how
much that
will feed
into other
and future

work
but it's
always very
exciting to
me when you
see different
fields at
least formally
being able to
be related to
each other
that usually
means there
something
something going
on
something
fundamental
we'll have
more to say
in that
way
so
and join
us
yeah
and
you're welcome
anytime
to participate
yeah
it'll be
Chris
and
Carl
and I think
Mike
I think

everybody's
coming
so if you
want to
come
and
join the
panel
you're
more than
welcome
I'll add
to the
events
and pop
in if
it works
so
thanks again
everybody
hope that
the viewers
can bear
with the
lag if it
was there
but
peace
out
everyone
thanks a lot
for this
great discussion
thanks for
having me
bye
bye
bye

thank you

thank you

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.